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Fabrication of Periodically-Inverted AlGaAs Waveguides for Quasi-Phase-Matched Wavelength Conversion at 1.55 μm

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We have improved fabrication processes of periodically inverted AlGaAs waveguides for highly efficient quasi-phase-matched wavelength conversion. In order to reduce propagation losses in waveguiding devices fabricated based on sublattice reversal epitaxy, we have introduced template planarization by chemical mechanical polishing and low-temperature regrowth technique. Corrugations at the core/clad interface and the propagation loss at 1.55 μm has been reduced to 40 nm and 9 dB/cm, respectively. First-order quasi-phase-matched second-harmonic generation was demonstrated at 1.55 μm fundamental wavelength with a 1.47-mm-long device. Obtained internal conversion efficiency is 2.2%/W which is one order of magnitude lower than the theoretical value for an ideal lossless waveguide. A significant increase in the conversion efficiency is expected with a redesign of the device and further optimization of the fabrication processes. © 2009 The Japan Society of Applied Physics

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1. Introduction

In the current telecommunication technology, wavelength division multiplexing (WDM) scheme is playing an important role. WDM enables the utilization of a significant portion of the available fiber bandwidth by allowing many independent signals to be transmitted simultaneously on one fiber, with each signal located at a different wavelength. To retain high-speed data transmission with WDM, the fast signal switching between various wavelengths is required. In order to overcome an optoelectronic switching speed bottleneck, all-optical wavelength conversion needs to be realized. Difference-frequency generation (DFG) offers a full range of transparency for all-optical wavelength conversion. Demands for the wavelength conversion devices include large bandwidths, high conversion efficiencies, robustness, and compactness. AlGaAs waveguiding devices have been investigated for this purpose because AlGaAs has a number of attractive properties. Since they exhibit much higher quadratic nonlinearities¹⁾ than conventional nonlinear optical crystals such as LiNbO₃, AlGaAs quasi phase matching (QPM) devices have a potential for realizing high conversion efficiencies with relatively short device lengths and/or lower pump intensities. Moreover, they can potentially be integrated with pumping laser diodes monolithically.

Since zincblende-type compound semiconductors have phase matching problem, i.e., birefringent phase matching is not achievable because of the optical isotropy of the cubic crystal system, some approaches have been investigated to achieve phase matching, such as artificial birefringence phase matching (BPM),^{2–4)} modal phase matching (MPM),^{5,6)} and QPM. QPM requires spatial modulation of the optical nonlinearities. Periodically inverted AlGaAs QPM waveguides based on direct wafer bonding followed by metal organic chemical vapor deposition (MOCVD) overgrowth have been fabricated and QPM second-harmonic generation (SHG)⁷⁾ and QPM-DFG⁸⁾ have been demonstrated in the wavelength range around 1.5 μm . Wafer bonding techniques, however, need complicated fabrication processes resulting in inevitable crystal defects in overgrown epitaxial

layers, which induce high waveguide losses. We have developed a crystal growth technique, sublattice-reversal epitaxy, for growing spatially inverted epitaxial GaAs layers.^{9–11)} Although several QPM-SHG experiments using the devices fabricated based on sublattice reversal epitaxy are demonstrated,^{11–13)} these reports have been suffered from severe propagation losses which hinder efficient nonlinear optical interactions over reasonably long lengths. Significantly high propagation losses in QPM AlGaAs waveguide are due to corrugations on the core/clad interfaces which are formed during the fabrication process. The first process resulting in the corrugation formation is preparation of the patterned templates which is periodically etched with QPM periods. The other factor for the corrugation formation is the regrowth process for fabricating AlGaAs waveguiding structures on QPM patterned GaAs templates, in which anisotropic surface migration of Ga atoms on GaAs(100) surfaces induces formation of corrugation.¹⁴⁾

In this paper, we report an improved process for fabricating AlGaAs QPM waveguides with suppressed corrugations. A result of a QPM-SHG experiment is also presented.

2. Device Fabrication

Figure 1 schematically shows an AlGaAs waveguiding device designed for QPM DFG in the 1.5 μm wavelength region pumpable with AlGaAs laser diodes oscillating around 0.78 μm . We adopted the ridge waveguide structure in order to suppress scattering losses due to side wall roughness. The ridge stripe is parallel to [011] upon which pump, signal and idler light waves interact via the nonlinear optical coefficient d_{14} , a single nonzero component of the nonlinear optical coefficient tensor of zincblende-type crystals.

We have fabricated QPM AlGaAs waveguides by using a standard solid-source molecular-beam epitaxy (MBE) based on sublattice-reversal epitaxy technique.¹⁰⁾ Outline of the fabrication process is as follows: 1) a spatially inverted GaAs layer is epitaxially grown on a Ge interlayer deposited on a GaAs(100) substrate misoriented toward [011] by 2°; 2) the GaAs epilayer and the Ge interlayer are periodically etched with the designed QPM period so that inverted and noninverted GaAs surfaces are alternately exposed; 3)

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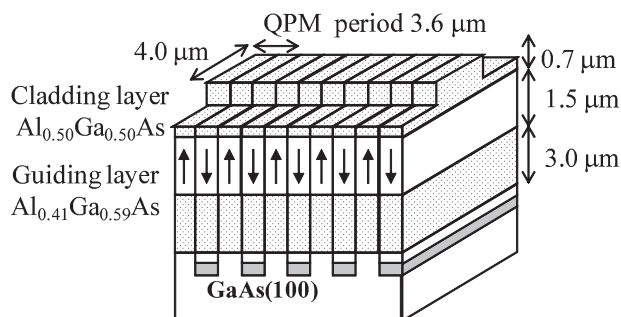


Fig. 1. Schematic drawing of the designed QPM-DFG AlGaAs waveguiding device.

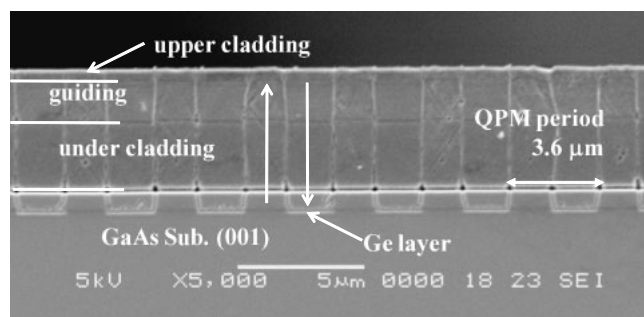


Fig. 3. Cross-sectional image of a stain-etched QPM AlGaAs waveguide fabricated using the improved process.

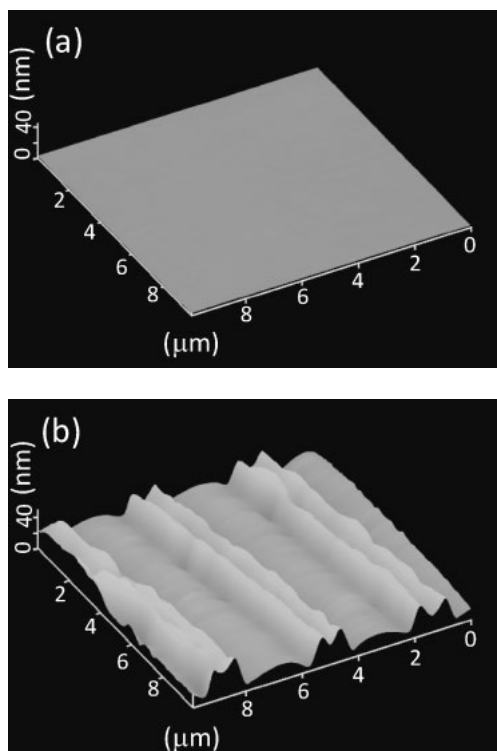


Fig. 2. AFM images of the surface of the GaAs patterned template planarized by CMP (a) and the surface of the fabricated AlGaAs waveguiding structures (b).

a thick GaAs epilayer is regrown on the template; 4) the corrugated surface of the regrown GaAs epilayer is planarized by chemical mechanical polishing (CMP); 5) AlGaAs clad/core/clad structure is formed on the planarized template by MBE regrowth; 6) 4-μm-wide ridges running along [011] are formed by wet etching.

In the present study, to suppress corrugation originating in the patterned GaAs template, we flattened the template surface by using CMP technique. We used 8 g/l NaOCl (Fujimi INSEC-NIB) as a slurry and a polyurethane pad (SURFIX) as a polishing cloth. Polishing pressure was reduced to as low as 60 g/cm² in order to obtain satisfactory smooth surfaces with smaller polishing depths. Figure 2(a) shows a surface profile of the obtained flattened patterned template measured using an atomic force microscope (AFM).

We also gave special attention to the process of AlGaAs regrowth for fabricating waveguiding structures. We have

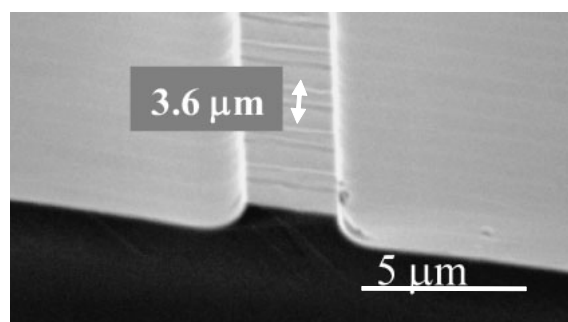


Fig. 4. Bird's-eye view of a ridge waveguide.

adopted the low-temperature growth technique in order to suppress the corrugation formation during the regrowth process. We have revealed that the corrugation formation is caused by the anisotropic surface migration of Ga atoms on GaAs(100) surfaces, and that the corrugation can be suppressed in low-temperature MBE regrowth.¹⁴⁾ We grew lower and upper cladding layers of Al_{0.5}Ga_{0.5}As at 400 °C, whereas we grew the guiding Al_{0.41}Ga_{0.59}As layer, where almost all the light powers are confined, under the standard high-temperature (520 °C) condition. The whole regrowth process was performed under a constant V/III ratio of 44, which is optimized for preserving QPM duty ratio. Surface profile of a fabricated AlGaAs waveguide is shown in Fig. 2(b). The surface corrugation height is reduced to as low as 40 nm. To the best of our knowledge, this is the lowest corrugation ever obtained in QPM AlGaAs waveguides. We have confirmed the residual corrugations are generated during the high-temperature growth of the guiding layers, and we believe that we will be able to further reduce the corrugations by optimizing the guiding layer growth conditions.

Figure 3 shows the cross-sectional scanning electron microscope (SEM) image of a stain-etched waveguide. Antiphase boundaries between inverted and noninverted domains propagate vertically preserving the QPM duty ratio, which has been made possible under the carefully controlled V/III ratio kept constant throughout the regrowth process. The bird's-eye view SEM image of a fabricated waveguide's ridge is shown in Fig. 4. Residual surface corrugations and slight regular sidewall corrugations have been observed. The sidewall corrugations might be caused by weak anisotropic nature of the wet etching we used in the ridge formation process.

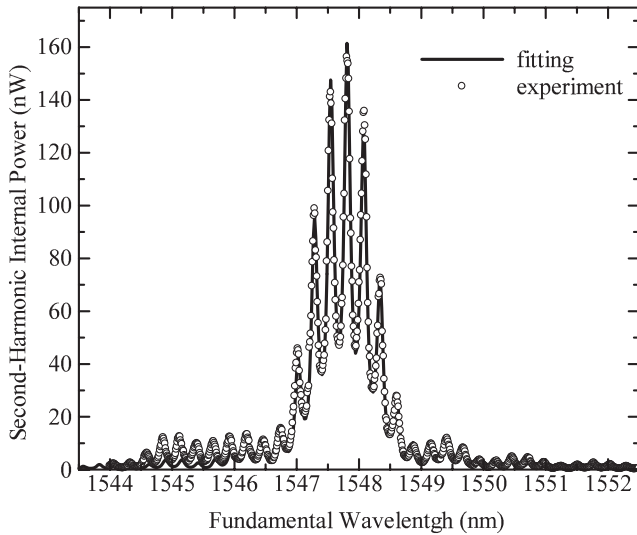


Fig. 5. Experimental and theoretical tuning curves of a 1.47-mm-long AlGaAs QPM waveguide.

3. Experimental Results

We characterized wavelength conversion performance of the fabricated devices by SHG pumped by a fundamental light source oscillating around 1.5 μm . The fundamental light source we used was a tunable external cavity InGaAsP laser diode (NEWFOCUS 6328-H). The transverse electric (TE)-polarized fundamental light was fed into the sample waveguide by end-fire coupling. Transverse magnetic (TM)-polarized second-harmonic output generated by the type-I QPM interaction was detected with a Si photodetector and the signal was amplified by using the standard lock-in technique. A 1.47-mm-long QPM waveguide was mounted on a temperature-stabilized holder. The SHG measurement was performed at room temperature.

The obtained second-harmonic power internal to the device is plotted as a function of the fundamental wavelength in Fig. 5. The first-order QPM is achieved at the fundamental wavelength of 1547.8 nm at room temperature. The coupled fundamental power internal to the device at the input facet is estimated to be 2.0 mW. The obtained maximum internal conversion efficiency is 3.9%/W.

The experimental data exhibit short-period deep oscillations superimposed on the sinc-squared tuning curve. The single-period oscillations are due to Fabry–Perot resonance of the fundamental lightwave in the cavity formed by the cleaved input and output facets. The clear Fabry–Perot oscillations indicate that the waveguide has quite low propagation loss at the fundamental wavelengths. However, no oscillatory characteristic due to the second-harmonic Fabry–Perot effect has been found, implying very high propagation loss for the second-harmonic lightwave. We plotted theoretical tuning curve taking account of Fabry–Perot enhancement of fundamental light and propagation losses based on the following equation:¹⁵⁾

$$P_{2\omega} = \eta^0 \exp\left[-\left(\alpha_\omega + \frac{\alpha_{2\omega}}{2}\right)L\right] \frac{\sinh^2\left[\left(\alpha_\omega - \frac{\alpha_{2\omega}}{2}\right)\frac{L}{2}\right]}{\left[\left(\alpha_\omega - \frac{\alpha_{2\omega}}{2}\right)\frac{L}{2}\right]^2}$$

$$\times \left[\frac{P_\omega^0}{1 - 2Re^{-\alpha_\omega L} \cos(4\pi n_{\text{eff}} L / \lambda) + R^2 e^{-2\alpha_\omega L}} \right]^2, \quad (1)$$

where P_ω^0 is the coupled fundamental light power internal to the device without Fabry–Perot enhancement measured at the input facet, $P_{2\omega}$ the internal second-harmonic power measured at the output facet, and λ , R , L , n_{eff} , α_ω , $\alpha_{2\omega}$ represent fundamental wavelength, facet power reflectivity, device length, effective index of the fundamental guided mode, fundamental propagation loss, and second-harmonic propagation loss, respectively. η^0 represents the normalized conversion efficiency for ideal lossless device without Fabry–Perot enhancement. In this equation, we neglected resonance effect for the second-harmonic wave because the Fabry–Perot resonance was not observed in the experiment. We measured fundamental propagation loss of the waveguide, α_ω , using standard Fabry–Perot technique. Measured propagation loss at 1.55 μm is 9 dB/cm which is much smaller than the previous value ~ 50 dB/cm obtained in devices with ~ 80 -nm-high corrugations fabricated by non-optimized processes. Propagation loss at the second-harmonic wavelength, $\alpha_{2\omega}$, was estimated by the fitting of the theoretical tuning curve given by eq. (1) to the experimental data. The estimated $\alpha_{2\omega}$ is ~ 100 dB/cm. As is clearly seen in Fig. 5, the experimental data are closely fitted with theoretical tuning curve indicating high accuracy of our experiment and high reliability of the determined parameters. The experimentally obtained normalized conversion efficiency η^0 is 820%/(W cm²). This is close to the theoretical value 1100%/(W cm²) which is predicted based on

$$\eta_{\text{theory}}^0 = \frac{8\pi^2}{\epsilon_0 c \lambda_\omega^2} \frac{d_{\text{eff}}^2}{n_\omega^2 n_{2\omega}} \frac{\left[\iint (E_\omega)^2 E_{2\omega} dx dy \right]^2}{\left(\iint E_\omega^2 dx dy \right)^2 \iint E_{2\omega}^2 dx dy}, \quad (2)$$

where d_{eff} , n_ω , $n_{2\omega}$, E_ω , and $E_{2\omega}$ represent effective nonlinear optical coefficient, effective refractive indices at the fundamental and the second-harmonic wavelengths, and electric field distributions of the fundamental and the second-harmonic waves, respectively. The conversion efficiency in the present device is evidently limited by large propagation losses: the obtained efficiency without the Fabry–Perot enhancement 2.2%/W is an order of magnitude lower than the theoretical one 23%/W. Although we have reduced the fundamental propagation loss to as low as 9 dB/cm, this is still significantly higher than that of blank waveguide fabricated under the same condition without periodically inverted structure (3 dB/cm). Without doubt, the excess loss is due to residual corrugation at the interface between the core and the upper cladding layers which is formed during the core layer regrowth process. Obviously, we have to further optimize the regrowth temperature. The extremely high loss at 0.78 μm is most likely due to an absorption in the band tail of the core material Al_{0.41}Ga_{0.59}As. Significant decreases in second-harmonic loss are expected with a redesign of the device with higher Al content of the core material.

4. Conclusions

We have improved fabrication process of periodically

inverted QPM AlGaAs waveguides. By introducing template planarization by CMP and low-temperature regrowth technique, the corrugation at the core/clad interface and the propagation loss at 1.55 μm has been reduced to 40 nm and 9 dB/cm, respectively. We have demonstrated first-order QPM SHG with 1.55 μm fundamental wavelength, and obtained an internal conversion efficiency of 2.2%/W for a 1.47-mm-long device, which is one order of magnitude lower than the theoretical value. Significant increases in conversion efficiency are expected with decreases in propagation losses at fundamental and second-harmonic wavelength regions. A redesign of the device with higher Al content for significant decrease in the second-harmonic loss and an optimization in the regrowth conditions for further reduction of corrugations are now underway.

Acknowledgments

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